

Asmita Pal

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Education

PhD in Computer Engineering

University of Wisconsin - Madison

Advisor : Joshua San Miguel

August 2018 - Expected May 2025

MS in Computer Engineering

Utah State University

Advisor : Koushik Chakraborty, Sanghamitra Roy

August 2016 - August 2018

B.Tech. in Electronics and Instrumentation Engineering

West Bengal University of Technology, India

July 2010 - June 2014

Publications

- **Asmita Pal**, Keerthana Desai, Rahul Chatterjee, Joshua San Miguel: **Camouflage: Utility-Aware Obfuscation for Accurate Simulation of Sensitive Program Traces**, published in ACM Trans. Archit. Code Optim. 21, 2, Article 36 (June 2024), 23 pages.
 - Mitali Soni, **Asmita Pal**, Joshua San Miguel: **As-Is Approximate Computing**, published in ACM Trans. Archit. Code Optim. 20, 1, Article 3 (March 2023), 26 pages.
 - **Asmita Pal**, Karthik Swaminathan, Subhankar Pal, Joshua San Miguel: **Characterizing Memory side channels in FHE applications** presented in DISCC Workshop (*held in conjunction with MICRO'22*)
 - Pramesh Pandey, **Asmita Pal**, Koushik Chakraborty, Sanghamitra Roy: **Reliability and Uniformity Enhancement in 8T-SRAM based PUFs operating at NTC**, published in IEEE/ACM International Symposium on Low Power Electronics and Design (ISLPED) 2018.
 - **Asmita Pal**, Aatreyi Bal, Koushik Chakraborty, Sanghamitra Roy: **Split Latency Allocator: Process Variation-Aware Register Access Latency Boost in a Near-Threshold Graphics Processing Unit**, published in J. Low Power Electronics 13(3): 419-427 (2017).
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Awards

- Winner of N+1 Institute Google Reverse Pitch Competition - (AI Sustainability) at UW-Madison.
 - Sarah and David Epstein Teaching Fellowship [Awarded by Department of Electrical and Computer Engineering, **UW-Madison**]
 - Student Research Grant [Awarded by Office of Diversity, Inclusion and Funding at **UW-Madison**]
 - Anita Borg Grace Hopper Celebration 2019 Scholarship
 - Wisconsin Distinguished Fellow - Schneider [Awarded by Department of Electrical and Computer Engineering, **UW-Madison**]
 - Rajendra Prasad and Yasoda Mani Grandhi Fellow [Awarded by Department of Electrical and Computer Engineering, **Utah State University**]
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Research Interests & Technical Skills

Research Interests : Computer Architecture, Approximate Computing, Privacy-preserving Traces, Fully Homomorphic Encryption, Network-on-Chips, Side-channel Attacks, Memory Access Patterns, ML for Systems

Programming Languages : C/C++, Python, Bash, Verilog

Architectural Tools : ChampSim, PinTool, Perf, likwid, Gem5, Sniper

Research Experience

University of Wisconsin - Madison

WISCONSIN, UNITED STATES

Graduate Research Assistant

August 2018 – Present

- Investigated timing-based side-channel leakage in Network-on-Chips (NoC) and applied temporal partitioning mechanisms to mitigate contention.
- Characterized memory side-channel attack models to analyze security vulnerabilities in applications encrypted using Fully Homomorphic Encryption.
- Designed privacy-preserving synthetic trace generation technique, leveraging program semantics, to enable trace sharing for confidential workloads.
- Analyzed memory access patterns using statistical and NLP models in the context of prefetching and replacement policies.
- Explored approximate computing paradigms to optimize workloads for energy efficiency.

Utah State University

UTAH, UNITED STATES

Graduate Research Assistant

August 2016 – July 2018

- Power-performance tradeoff analysis for modern GPGPU architectures, focusing on register file optimizations.
- Investigated multi-core performance bottlenecks using thread criticality prediction.
- Analyzed the role of Deep Neural Networks in image recognition and speaker identification.

Industry Experience

AMD Research

BOSTON, UNITED STATES

Research Intern

June 2022 – Sept 2022

- Developed last level cache (LLC) replacement policies based on data reuse, to optimize performance for AI and HPC workloads, in SoC-based architectures.

AMD Research

BOSTON, UNITED STATES

Research Intern

September 2020 – December 2020

- Profiled memory-bound AI applications, to identify acceleration opportunities and enable more efficient data flow.

Infosys Technologies Limited

PUNE, INDIA

System Engineer

July 2014 – August 2016

- Developed an automation framework automation frameworks for hardware verification, streamlining debugging processes and reducing manual verification time, to enhance team productivity.
- Managed client deadlines and service delivery, coordinating with cross-functional teams.

Projects

Side-channel attacks for mesh-based Network-on-Chips(NOC)

Dec 2023 – Present

Supervisor : Joshua San Miguel

- Identified NoC contention issues in mesh-based interconnects.
- Developed temporal partitioning techniques to mitigate security vulnerabilities in SoCs.

Characterizing Memory Side Channels of FHE Applications

Dec 2021 – Dec 2023

Supervisor : Joshua San Miguel

- Analyzed side-channels in applications encrypted using FHE (Fully Homomorphic Encryption), on the IBM HELayers framework.
- Studied memory access patterns for a database encrypted query and classified different queries.

Camouflage : Privacy-preserving Traces

Jan 2020 – Oct 2023

Supervisor : Joshua San Miguel

- Characterized inherent architectural properties of program structures that influence program behavior, as captured in memory traces.
- Proposed a trace-based obfuscation mechanism based on program semantics, with a security-conservative mindset.
- Developed a threat model based on input indistinguishability to ensure input security.
- Tested AES traces against cache-based side-channel attacks.

As-Is Approximate Computing

Jan 2019 – Nov 2021

Supervisor : Joshua San Miguel

- Characterized applications for an approximate compute model with interruptive execution.
- Simulated approximate applications on a speculation-based architectural simulator for studying accuracy-runtime trade-offs.

Analyzing patterns in memory accesses for Prefetching

Sep 2018 – Dec 2019

Supervisor : Joshua San Miguel

- Applied natural language processing(NLP) algorithms to memory access sequences for address prediction.
- Verified hardware prefetcher vs NLP-predicted addresses to understand prefetcher accuracy and coverage.

Digit Recognition on FPGA using Multilayer Perceptron

Sep 2019 – Dec 2019

Supervisor : Parmesh Ramanathan

- Designed processor, memory and accelerator modules to enable storage of a pre-trained model and offload computations to accelerator.
- Captured image and predicted image digit on the 7 segment display.

Precise encoding in Stochastic Computing for Neural Networks

Sep 2018 – Dec 2018

Supervisor : Younghyun Kim

- Proposed a new encoding scheme for Stochastic Numbers and implemented this new encoding format in MAC module of a small-scale Neural Network.
- Analyzed the energy-accuracy trade-off for LeNet300-100 and observed up to 5X energy improvement.

Design of a Process-Variation Aware Low Power GPU

Jan 2017 – July 2017

Supervisor : Koushik Chakraborty, Sanghamitra Roy

- Explored the role of process variation on register files for a GPU operating at Near Threshold Voltages.
- Analyzed the performance variation in GPGPU applications from AMD APP SDK suite, with varying register latency.
- Proposed a low overhead technique to overcome the effects of process variation on register latency. Exhibited ~36% reduction in GPU energy consumption with marginal area and power overheads.

Speaker Diarization using Deep Neural Networks

Sep 2017 – Dec 2017

Supervisor : Todd Moon

- Extracted features from a conversation and used Deep Neural Networks to separate an audio signal according to speaker identity.
- Compared accuracy of the speaker diarization framework, using different deep neural networks (DNN) and convolutional neural networks (CNN).

Critical Thread Predictor

Jan 2017 – April 2017

Supervisor : Koushik Chakraborty

- Studied the statistical correlation of thread criticality metrics based on cache-misses and synchronization bottlenecks.
- Analyzed performance numbers for Splash2 Benchmarks on multi-core Sniper simulator.

GPU Power and Area Analysis

Jan 2017 – April 2017

Supervisor : Sanghamitra Roy

- Synthesis, place and route for basic pipe stages of a GPU, to study of change in power and area and timing violation, with frequency and technology node variation.

Design of a Global Router

Aug 2016 – Dec 2016

Supervisor : Sanghamitra Roy

- Developed a VLSI floorplan based on simulated annealing algorithm.
- Implemented a global routing engine using A* search algorithm

Professional Service

Invited Talks

- Town Hall - University of Wisconsin - Madison, Dept. of Electrical and Computer Engineering, February 2025
- Computer Architecture Coordination Seminar Series - University of Central Florida, November 2023

Artifact Evaluation Committee

- IEEE/ACM International Symposium on Microarchitecture 2023 (MICRO 2023, 2022)
- Architectural Support for Programming Languages and Operating Systems (ASPLOS 2022)
- Conference on Machine Learning and Systems (MLSys 2020)
- Architectural Support for Programming Languages and Operating Systems (ASPLOS 2020)

Teaching

ECE 757 (Advanced Computer Architecture II)

Spring 2024, Spring 2025

ECE 352 (Digital System Fundamentals)

Fall 2024

ECE 5750 (High Performance Computer Architecture)

Fall 2017